

# Mathematics: Geometry

## Course Information

**Course Name:** Mathematics – Geometry

**Level:** Beginner / Intermediate

**Prerequisites:** Basic arithmetic and elementary mathematics

**Estimated Duration:** 8–10 Hours

### Course Description

Geometry is a branch of mathematics that studies shapes, sizes, positions, angles, distances, and properties of space. In this course, students will learn fundamental geometric concepts including points, lines, angles, triangles, quadrilaterals, polygons, circles, perimeter, area, volume, coordinate geometry, and the Pythagorean theorem. Students will also solve practical problems involving geometric shapes and measurements.

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## Learning Outcomes

After completing this course, students will be able to:

- Define basic geometric terms.
  - Identify points, lines, line segments, and rays.
  - Classify and measure angles.
  - Identify different types of triangles.
  - Apply properties of triangles.
  - Understand quadrilaterals and polygons.
  - Calculate perimeter and area.
  - Understand circles and their properties.
  - Calculate circumference and area of circles.
  - Apply the Pythagorean theorem.
  - Calculate surface area and volume of three-dimensional shapes.
  - Understand basic coordinate geometry.
  - Solve practical geometry problems.
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# Lesson 1: Introduction to Geometry

## Learning Objectives

After this lesson, students will be able to:

- Define geometry.
  - Identify basic geometric objects.
  - Understand dimensions in geometry.
  - Distinguish between 2D and 3D shapes.
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## What is Geometry?

**Geometry** is the branch of mathematics that studies shapes, sizes, positions, angles, and properties of space.

Geometry is used in:

- Architecture
  - Engineering
  - Construction
  - Computer graphics
  - Design
  - Physics
  - Navigation
  - Manufacturing
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## Dimensions

### One-Dimensional

A one-dimensional object has length but no width or height.

Example:

Line

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### Two-Dimensional

A two-dimensional shape has length and width.

Examples:

- Square
  - Rectangle
  - Triangle
  - Circle
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### Three-Dimensional

A three-dimensional object has length, width, and height.

Examples:

- Cube
  - Cuboid
  - Cylinder
  - Sphere
  - Cone
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## Lesson 2: Points, Lines, and Line Segments

### Point

A point represents an exact position in space.

It is usually represented using a capital letter.

Example:

A •

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### Line

A line extends infinitely in both directions.

It can be represented as:



A line containing points A and B can be written as:

$\leftrightarrow AB$

## Line Segment

A line segment has two endpoints.

A •————• B

It has a definite length.

## Ray

A ray has one endpoint and extends infinitely in one direction.

A •————→

## Important Difference

| Object       | Endpoints | Length    |
|--------------|-----------|-----------|
| Point        | 0         | No length |
| Line         | 0         | Infinite  |
| Line Segment | 2         | Fixed     |
| Ray          | 1         | Infinite  |

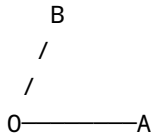
## Lesson 3: Angles

### What is an Angle?

An angle is formed when two rays meet at a common endpoint.

The common endpoint is called the **vertex**.

Example:



The angle can be written as:

$\angle BOA$

## Measuring Angles

Angles are usually measured in **degrees**.

The symbol for degrees is:

$^{\circ}$

## Lesson 4: Types of Angles

### Acute Angle

An acute angle is greater than  $0^{\circ}$  and less than  $90^{\circ}$ .

Example:

$45^{\circ}$

### Right Angle

A right angle measures exactly:

$90^{\circ}$

It is often represented by a small square.

## Obtuse Angle

An obtuse angle is greater than  $90^\circ$  and less than  $180^\circ$ .

Example:

$120^\circ$

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## Straight Angle

A straight angle measures:

$180^\circ$

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## Reflex Angle

A reflex angle is greater than  $180^\circ$  and less than  $360^\circ$ .

Example:

$270^\circ$

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## Complete Angle

A complete angle measures:

$360^\circ$

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# Lesson 5: Angle Relationships

## Complementary Angles

Two angles are complementary if their sum is:

$90^\circ$

Example:

$$30^\circ + 60^\circ = 90^\circ$$

Therefore,  $30^\circ$  and  $60^\circ$  are complementary angles.

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## Supplementary Angles

Two angles are supplementary if their sum is:

$$180^\circ$$

Example:

$$110^\circ + 70^\circ = 180^\circ$$

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## Vertically Opposite Angles

When two lines intersect, opposite angles are equal.

If one angle is:

$$60^\circ$$

the vertically opposite angle is also:

$$60^\circ$$

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# Lesson 6: Triangles

## What is a Triangle?

A triangle is a polygon with:

- 3 sides
- 3 vertices
- 3 angles

The sum of the interior angles of a triangle is:

180°

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## Lesson 7: Types of Triangles by Sides

### Equilateral Triangle

All three sides are equal.

If each angle is measured:

60°

then:

$$60^\circ + 60^\circ + 60^\circ = 180^\circ$$

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### Isosceles Triangle

An isosceles triangle has two equal sides.

The angles opposite the equal sides are also equal.

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### Scalene Triangle

A scalene triangle has three different side lengths.

Its angles are generally different as well.

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## Lesson 8: Types of Triangles by Angles

### Acute Triangle

All three angles are less than 90°.

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### Right Triangle

One angle is exactly 90°.



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## Obtuse Triangle

One angle is greater than  $90^\circ$ .

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## Lesson 9: Triangle Formulas

### Perimeter

The perimeter of a triangle is:

$$P = a + b + c$$

Example:

If the sides are:

5 cm, 6 cm, 7 cm

then:

$$\begin{aligned} P &= 5 + 6 + 7 \\ P &= 18 \text{ cm} \end{aligned}$$

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### Area

The area of a triangle is:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

Example:

Base = 10 cm

Height = 6 cm

$$\begin{aligned} \text{Area} &= \frac{1}{2} \times 10 \times 6 \\ \text{Area} &= 30 \text{ cm}^2 \end{aligned}$$

## Lesson 10: Pythagorean Theorem

The Pythagorean theorem applies to right-angled triangles.

The formula is:

$$a^2 + b^2 = c^2$$

where:

- $a$  and  $b$  are the legs.
- $c$  is the hypotenuse.

The hypotenuse is the side opposite the right angle.

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### Example

Suppose:

$$\begin{aligned} a &= 3 \\ b &= 4 \end{aligned}$$

Then:

$$3^2 + 4^2 = c^2$$

$$9 + 16 = c^2$$

$$25 = c^2$$

Therefore:

$$c = 5$$

So the triangle has sides:

$$3, 4, 5$$

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# Lesson 11: Quadrilaterals

A **quadrilateral** is a polygon with four sides.

Examples include:

- Square
- Rectangle
- Parallelogram
- Rhombus
- Trapezium
- Kite

The sum of the interior angles of a quadrilateral is:

360°

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# Lesson 12: Square

A square has:

- 4 equal sides
  - 4 right angles
  - Opposite sides parallel
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## Perimeter of Square

$$P = 4s$$

where  $s$  is the side length.

Example:

If:

$$s = 5 \text{ cm}$$

then:

$$P = 4 \times 5$$

$$P = 20 \text{ cm}$$

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## Area of Square

$$A = s^2$$

Example:

$$A = 5^2$$

$$A = 25 \text{ cm}^2$$

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## Lesson 13: Rectangle

A rectangle has:

- Opposite sides equal
- Four right angles
- Opposite sides parallel

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## Perimeter

$$P = 2(l + w)$$

where:

- $l$  = length
- $w$  = width

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## Area

$$A = l \times w$$

Example:

Length = 8 cm

Width = 5 cm

$$A = 8 \times 5$$
$$A = 40 \text{ cm}^2$$

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## Lesson 14: Parallelogram

A parallelogram has:

- Opposite sides parallel
- Opposite sides equal
- Opposite angles equal

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### Area

$$A = \text{base} \times \text{height}$$

Example:

Base = 10 cm

Height = 4 cm

$$A = 10 \times 4$$
$$A = 40 \text{ cm}^2$$

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## Lesson 15: Rhombus

A rhombus has four equal sides.

Its opposite sides are parallel.

The diagonals of a rhombus intersect at right angles.

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### Area

The area can be calculated using the diagonals:

$$A = 1/2 \times d_1 \times d_2$$

Example:

$$d_1 = 8 \text{ cm}$$

$$d_2 = 6 \text{ cm}$$

Then:

$$A = 1/2 \times 8 \times 6$$

$$A = 24 \text{ cm}^2$$

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## Lesson 16: Trapezium

A trapezium is a quadrilateral with one pair of parallel sides.

The parallel sides are commonly called the bases.

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### Area

$$A = 1/2 \times (a + b) \times h$$

where:

- **a** = first parallel side
- **b** = second parallel side
- **h** = height

Example:

$$a = 8 \text{ cm}$$

$$b = 12 \text{ cm}$$

$$h = 5 \text{ cm}$$

Therefore:

$$A = 1/2 \times (8 + 12) \times 5$$

$$A = 50 \text{ cm}^2$$

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## Lesson 17: Polygons

A **polygon** is a closed two-dimensional shape made from straight line segments.

Examples:

| Number of Sides | Name          |
|-----------------|---------------|
| 3               | Triangle      |
| 4               | Quadrilateral |
| 5               | Pentagon      |
| 6               | Hexagon       |
| 7               | Heptagon      |
| 8               | Octagon       |
| 9               | Nonagon       |
| 10              | Decagon       |

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### Interior Angle Sum

The sum of the interior angles of an -sided polygon is:

$$(n - 2) \times 180^\circ$$

Example:

For a pentagon:

$$\begin{aligned}(5 - 2) \times 180^\circ \\&= 3 \times 180^\circ \\&= 540^\circ\end{aligned}$$

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## Lesson 18: Circles

A circle is a two-dimensional shape consisting of all points that are the same distance from a fixed point.

The fixed point is called the **center**.

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## Important Parts of a Circle

### Radius

The distance from the center to any point on the circle.

It is represented by:

$r$

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### Diameter

The distance across the circle through its center.

$d = 2r$

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### Circumference

The distance around the circle.

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## Lesson 19: Circle Formulas

### Circumference

$C = 2\pi r$

or:

$C = \pi d$

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### Area

$A = \pi r^2$

Use:



$$\pi \approx 3.14159$$

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## Example

If:

$$r = 5 \text{ cm}$$

Area:

$$A = \pi \times 5^2$$

$$A = 25\pi$$

Approximately:

$$A \approx 78.54 \text{ cm}^2$$

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## Lesson 20: Three-Dimensional Shapes

Three-dimensional shapes have:

- Length
- Width
- Height

Examples:

- Cube
  - Cuboid
  - Cylinder
  - Cone
  - Sphere
- 

## Lesson 21: Cube

A cube has six equal square faces.

If the side length is  $s$ :

## Volume

$$V = s^3$$

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## Surface Area

$$SA = 6s^2$$

Example:

If:

$$s = 4 \text{ cm}$$

Volume:

$$V = 4^3$$
$$V = 64 \text{ cm}^3$$

Surface area:

$$SA = 6 \times 4^2$$
$$SA = 96 \text{ cm}^2$$

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## Lesson 22: Cuboid

A cuboid has:

- Length
- Width
- Height

## Volume

$$V = l \times w \times h$$

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## Surface Area

$$SA = 2(lw + lh + wh)$$

Example:

$$\begin{aligned}l &= 5 \text{ cm} \\w &= 3 \text{ cm} \\h &= 2 \text{ cm}\end{aligned}$$

Volume:

$$\begin{aligned}V &= 5 \times 3 \times 2 \\V &= 30 \text{ cm}^3\end{aligned}$$

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## Lesson 23: Cylinder

A cylinder has two circular bases.

### Volume

$$V = \pi r^2 h$$

where:

- $r$  = radius
- $h$  = height

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### Curved Surface Area

$$CSA = 2\pi rh$$

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### Total Surface Area

$$TSA = 2\pi r(r + h)$$

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## Lesson 24: Sphere

A sphere is a three-dimensional object in which every point on its surface is the same distance from its center.

### Volume

$$V = \frac{4}{3} \pi r^3$$

### Surface Area

$$SA = 4\pi r^2$$

## Lesson 25: Cone

A cone has:

- One circular base
- One curved surface
- One vertex

### Volume

$$V = \frac{1}{3} \pi r^2 h$$

where:

- $r$  = radius
- $h$  = height

## Lesson 26: Coordinate Geometry

Coordinate geometry combines algebra and geometry.

A point on a two-dimensional coordinate plane is represented as:

$$(x, y)$$

The coordinate system contains:

- x-axis
- y-axis
- Origin

The origin is:

(0, 0)

## Quadrants

The coordinate plane has four quadrants.

| Quadrant | x | y |
|----------|---|---|
| I        | + | + |
| II       | - | + |
| III      | - | - |
| IV       | + | - |

## Lesson 27: Distance Between Two Points

For two points:

$A(x_1, y_1)$   
 $B(x_2, y_2)$

the distance is:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

## Example

Consider:

A(1,2)  
B(4,6)

Then:

$$d = \sqrt{[(4-1)^2 + (6-2)^2]}$$

$$d = \sqrt{3^2 + 4^2}$$

$$d = \sqrt{25}$$

Therefore:

$$d = 5$$

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## Lesson 28: Midpoint Formula

The midpoint of two points:

A( $x_1, y_1$ )  
B( $x_2, y_2$ )

is:

$$M = ((x_1+x_2)/2, (y_1+y_2)/2)$$

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### Example

Given:

A(2,4)  
B(6,8)

The midpoint is:

$$M = ((2+6)/2, (4+8)/2)$$

Therefore:

$$M = (4, 6)$$

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## Lesson 29: Transformations

A transformation changes the position or orientation of a shape.

Common transformations include:

- Translation
- Reflection
- Rotation
- Enlargement

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### Translation

A translation moves a shape without changing its size or orientation.

Example:

$$(x, y) \rightarrow (x+3, y+2)$$

This moves a point 3 units right and 2 units up.

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### Reflection

Reflection creates a mirror image.

For reflection across the x-axis:

$$(x, y) \rightarrow (x, -y)$$

For reflection across the y-axis:

$$(x, y) \rightarrow (-x, y)$$

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## Rotation

Rotation turns a shape around a fixed point.

A  $90^\circ$  counterclockwise rotation around the origin is:

$$(x, y) \rightarrow (-y, x)$$

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## Lesson 30: Similarity and Congruence

### Congruent Shapes

Two shapes are congruent if they have the same:

- Shape
- Size

They may be rotated or reflected.

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### Similar Shapes

Two shapes are similar if they have the same shape but may have different sizes.

Corresponding angles are equal, and corresponding sides are proportional.

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## Solved Problems

### Problem 1: Triangle Area

A triangle has a base of 12 cm and height of 8 cm.

Find its area.

#### Solution

$$A = \frac{1}{2} \times \text{base} \times \text{height}$$

$$A = \frac{1}{2} \times 12 \times 8$$



$$A = 48 \text{ cm}^2$$

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## Problem 2: Rectangle Area

A rectangle has length 10 cm and width 6 cm.

Find its area.

### Solution

$$A = l \times w$$

$$A = 10 \times 6$$

$$A = 60 \text{ cm}^2$$

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## Problem 3: Circle Circumference

A circle has a radius of 7 cm.

Find its circumference.

### Solution

$$C = 2\pi r$$

$$C = 2 \times \pi \times 7$$

Using:

$$\pi \approx 3.14159$$

$$C \approx 43.98 \text{ cm}$$

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## Problem 4: Pythagorean Theorem

A right triangle has sides 6 cm and 8 cm.

Find the hypotenuse.

### Solution

$$a^2 + b^2 = c^2$$

$$6^2 + 8^2 = c^2$$

$$36 + 64 = c^2$$

$$100 = c^2$$

Therefore:

$$c = 10 \text{ cm}$$

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## Problem 5: Cuboid Volume

A cuboid has:

Length = 8 cm  
Width = 5 cm  
Height = 3 cm

Find the volume.

### Solution

$$V = l \times w \times h$$

$$V = 8 \times 5 \times 3$$

$$V = 120 \text{ cm}^3$$

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# Practice Exercises

## Exercise 1

Classify the following angles:

1.  $35^\circ$
  2.  $90^\circ$
  3.  $125^\circ$
  4.  $180^\circ$
  5.  $270^\circ$
- 

## Exercise 2

A triangle has sides:

7 cm, 8 cm, 9 cm

Find its perimeter.

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## Exercise 3

A rectangle has:

Length = 15 cm  
Width = 8 cm

Find:

1. Area
  2. Perimeter
- 

## Exercise 4

A square has a side length of 12 cm.

Find:

1. Area
  2. Perimeter
-

## Exercise 5

A circle has a radius of 10 cm.

Find:

1. Diameter
  2. Circumference
  3. Area
- 

## Exercise 6

A cube has a side length of 5 cm.

Find:

1. Volume
  2. Surface area
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## Exercise 7

Find the distance between:

$A(2, 3)$   
 $B(6, 6)$

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## Exercise 8

Find the midpoint of:

$A(2, 4)$   
 $B(8, 10)$

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## Mini Project

### Designing a Small Park

Design a small rectangular park using geometric concepts.

The park contains:

- A rectangular main area
- A circular fountain
- A triangular garden
- A square playground

Students should calculate:

1. Perimeter of the park.
2. Area of the park.
3. Area of the fountain.
4. Area of the triangular garden.
5. Area of the playground.
6. Total area occupied by all sections.
7. Remaining open area.

Students should draw a simple diagram showing the different shapes.

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## Final Quiz

1. What is geometry?
2. What is a point?
3. What is a line?
4. What is a line segment?
5. What is a ray?
6. What is an angle?
7. What is an acute angle?
8. What is a right angle?
9. What is an obtuse angle?
10. What are complementary angles?
11. What are supplementary angles?
12. What is the sum of the angles in a triangle?
13. What is an equilateral triangle?
14. What is an isosceles triangle?
15. What is a scalene triangle?
16. What is the Pythagorean theorem?
17. What is a quadrilateral?
18. What is the formula for the area of a rectangle?
19. What is the formula for the area of a triangle?
20. What is the formula for the area of a circle?
21. What is the difference between radius and diameter?
22. What is the formula for the volume of a cube?
23. What is the formula for the volume of a cylinder?
24. What is coordinate geometry?
25. What is the distance formula?
26. What is the midpoint formula?
27. What are congruent shapes?
28. What are similar shapes?

29. What is a transformation?  
30. Give two real-world applications of geometry.
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## Course Summary

After completing this course, students should understand the fundamental concepts of geometry and be able to solve basic and intermediate geometric problems.

The major concepts covered are:

- Introduction to geometry
- Points
- Lines
- Line segments
- Rays
- Angles
- Angle relationships
- Triangles
- Triangle properties
- Pythagorean theorem
- Quadrilaterals
- Squares
- Rectangles
- Parallelograms
- Rhombuses
- Trapeziums
- Polygons
- Circles
- Perimeter
- Area
- Surface area
- Volume
- Three-dimensional shapes
- Coordinate geometry
- Distance formula
- Midpoint formula
- Transformations
- Similarity
- Congruence
- Practical geometry problems

## Next Step

After completing Geometry, students can continue with **Algebra, Trigonometry, Coordinate Geometry, Probability, or Basic Statistics.**